# Choose a reconstruction by its conditional squared-error risk

**Choose a reconstruction after observing $y$**

$$m = E[z|y]$$

Unknown $z \in \mathbb{R}^d$; choose $a \in \mathbb{R}^d$ to reconstruct it.

**Squared-error criterion**

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

- Choose $a$ to minimize conditional expected squared error $R(a|y)$.

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$$r = z - m, \quad z - a = r + (m - a)$$

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$$\text{align(center)}[\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2]$$

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